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Bridging ideologies: analyzing the use of moral language and framing in social media discourse on climate change by U.S. congress members through computational approaches

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Abstract
The discourse on climate change transcends scientific discussions and policy debates, often incorporating moral language and ethical considerations. This study explores framing strategies in political persuasion and the underlying moral foundations associated with climate change by combining computational methods and critical discourse analysis of tweets from 111th-117th U.S. Congresses members. The aim is to map out the bipartisan trends in the use of moral language and framing concerning climate change issues, thereby enriching the understanding of public opinion dynamics and the evolving partisan divide on climate action. Our findings reveal an intensifying partisan polarization in framing. Contrary to the expected moral divide, we uncover a bipartisan agreement on the moral foundation of care and fairness, and a consistent cross-party employment of moral language associated with frames over time. The interplay between generic frames and moral foundations suggests the potential for collective action on climate change across the political spectrum.

Keywords Climate change · U.S. Congress · Framing · Moral foundations theory · Computational methods

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The discourse surrounding climate change is not only a matter of scientific consensus and policy debate but is also deeply intertwined with moral and ethical considerations (Markowitz and Shariff 2012). As the global community grapples with the escalating impacts of climate change, the moral dimension of environmental stewardship has become increasingly prominent. This intersection of morality and climate discourse has garnered significant academic interest, leading researchers to explore how moral concerns are articulated in the context of climate change and how these articulations influence public perception and policy preferences.

Moral language cues have been identified as an important factor in guiding individuals' attention (Brady et al. 2019). Moral language, and the ways that language is used to frame issues along moral lines, can activate and intensify beliefs, and change the types of decisions and policies people are willing to accept (Anderson et al. 2013). Furthermore, as moral values are associated with social and political identity and represent politicized discourse (Wetts 2020), the use of moral frames can trigger in-group identification and loyalty, thereby yielding stronger persuasive effects (Hurst and Stern 2020). A pivotal area of study within this realm focuses on the moral frameworks employed by political elites to reinforce solidarity within their respective political factions. Existing research (e.g., Graham et al. 2009; Sterling and Jost 2018) highlights how moral language as a vehicle for conveying moral values plays a crucial role in the persuasive tactics of political figures aiming to galvanize their base.

However, the moral divide between conservatives and liberals is by no means absolute (Dickinson et al. 2016) or binary (Graham et al. 2009). Sterling and Jost (2018) found in an analysis of congressional tweets from 2014 that Democrats, in addition to using moral values of care and fairness ("individualizing foundations"), also employed purity and loyalty-related language ("binding foundations"), which is assumed to be more commonly used by Republicans. Day et al. (2014) found that when general pro-environmental appeals matched the moral values of conservatives, they had consistent positive effects across an array of environmental attitudes and behaviors (see also Wolsko et al. 2016). More broadly, recent research found no evidence for a consistent pattern of ideological differences in the use of moral language across topics (Wang and Inbar 2021).

The debate over climate change constitutes an illuminating case for the complex strategies politicians employ to shape public opinion and policy. Central to this dialogue are the notions of generic frames and moral language, which serve as critical tools in the arsenal of political communication. Generic frames offer a lens through which complex issues are simplified and made relatable to the public, facilitating engagement with and understanding of the climate crisis. Moral language, on the other hand, appeals to the ethical values and principles of individuals, evoking emotional responses and moral judgments that can significantly sway public opinion and drive policy initiatives. For example, Vainio and Mäkiniemi (2016) found that there were differences between general and issue-specific moral concerns, which have implications for their association with political orientation and pro-environmental behavior. The endorsement of general moral foundations made the effect of political orientation on climate-friendly choices disappear. They further developed an issue-specific measure of moral foundations and found that binding foundations (i.e., ingroup/loyalty, authority/respect, and purity/sanctity) were associated with the avoidance of climate-friendly choices among right-wing individuals.

This study explores the moral foundations and framing strategies used in political persuasion regarding climate change in the U.S., where the issue remains highly politicized. It underscores the divergent approaches stemming from broader ideological belief systems and their alignment with political parties in the U.S. Specifically, the study highlights that political liberals, who tend to align with the Democratic Party, generally acknowledge and advocate for policies addressing anthropogenic climate change, whereas political conservatives, who are more often associated with the Republican Party, have demonstrated significant skepticism and resistance to such policies over the last two decades (Andrews and Smirnov 2020; McCright and Dunlap 2011). While this correlation is not absolute—acknowledging that not all Democrats are liberal and not all Republicans are conservative—it reflects a prevailing trend in U.S. politics.[1] This division is sometimes attributed to different moral foundations that underpin the political spectrum (e.g., Park et al. 2010). Drawing on Moral Foundations Theory (MFT), previous studies has shown that the political left tends to emphasize values of fairness and care ("individualizing foundations"), whereas the right prioritizes loyalty, sanctity, and authority ("binding foundations") (e.g., Kugler et al. 2014). This dichotomy in moral prioritization not only shapes political identities but also informs the strategies employed in political messaging. Such targeted communication, adopting the specific moral languages resonant with these foundational values, has been demonstrated to effectively engage and motivate policy compliance across the ideological spectrum (cf. Ryan 2014; Salomon et al. 2017; Skitka and Bauman 2008).

The objective of this study is to map out the bipartisan trends in moral language and framing of climate change issues, thereby enriching the understanding of public opinion dynamics and the evolving partisan divide on climate action. The distinction in moral language and framing between liberals and conservatives points to a deeper ideological divide in perceptions of environmental responsibility and ethical obligation. It underscores the role of moral foundations in shaping political discourse on climate change, suggesting that effective communication strategies may need to account for these underlying moral differences.

Moreover, our analysis aims to uncover common concerns and moral foundations regarding climate change that transcend party lines. By understanding the moral underpinnings of climate discourse, policymakers, activists, and communicators can craft more resonant messages that bridge ideological divides and foster a more inclusive approach to addressing the global climate crisis. There is a heightened need for research to explore the nuanced relationship between moral language, socioeconomic themes, and climate change discourse. Such investigations could provide invaluable insights into how moral foundations influence framing strategies and, in turn, public engagement with climate issues.

Our study employs automatic text analysis of social media posts—specifically, tweets about climate change made by members of the 111th to 117th U.S. Congress. Given Twitter's (now known as X) pivotal role in online political discourse, this platform serves as a primary medium for Congress members to engage with constituents and the public at large (Jones et al. 2018). Due to the Congress members' status as public figures, we argue that their tweets go beyond mere expressions of personally held beliefs or attitudes to strategi-

[1] In the United States, political party affiliation often correlates with ideological leanings, but significant diversity exists within each party. The Democratic Party is generally associated with more liberal positions, particularly on issues like climate change, while the Republican Party is commonly linked to more conservative stances. However, not all Democrats identify as liberals, nor do all Republicans identify as conservatives. These correlations reflect broad trends rather than rigid definitions.

cally align with their partisan interests and their constituents' expectations, even though their expressions may be conveyed in a manner that evokes a sense of authenticity. Methodologically, we first utilized the extended Moral Foundations Dictionary to assess the moral content of Congress members' tweets, enhancing the precision of moral foundation measurements through crowd-sourced lexicon annotations. Next, we employed a hybrid computer-assisted method to identify and classify the underlying climate change frames in these tweets, integrating corpus linguistics with theory-driven deductive coding to outline the discourse strategies of Congress members. By utilizing computer-assisted text analysis techniques, we aim to provide a nuanced perspective on the communicative dynamics underpinning the discourse on climate change among members of Congress (Caprara et al. 2006; Graham et al. 2009; Neiman et al. 2016; Sagi and Dehghani 2014).

1 Theoretical background

1.1 Moral foundations, political persuasion and environmental issues

The discourse on climate change often intersects with both moral and material considerations, underscoring the complexity of persuasion strategies employed by political actors (Wolsko et al. 2016). Two independent yet interrelated dimensions of moral foundation expressions are worth noting: moral values and moral language use. Moral values justify ingroup beliefs with sacred principles, resulting in different usage of moral language to defend the group identity and outcast "wicked" outgroup members (Wang and Inbar 2021). Moral language is a tool often used by politicians to influence public debates and persuade voters (Clifford and Jerit 2013). Since voters tend to associate an issue with their moral values, morality can then function as a motivator for actions (Fiske and Rai 2014).

On the issue of climate change specifically, moralization is arguably a cause for the widespread political polarization (Feinberg and Willer 2013). Liberals are more likely to view climate change as a moral issue than conservatives, and are more likely to consider a range of specific pro-environmental behaviors (e.g., recycling, buying green products, and using reusable bags) to be "moral" (Currie and Choma 2018). To closely examine the moral communication surrounding climate change, we first review existing theories of moral values, and then explore the scholarship on moral language use.

One of the most influential theories concerning moral values is the Moral Foundations Theory (MFT), a comprehensive framework that delineates fundamental moral values universally recognized across cultures. The MFT identifies five core moral domains: harm/care, fairness/reciprocity, ingroup/loyalty, authority/respect, and purity/sanctity. Of these, harm/care and fairness/reciprocity are categorized as individualizing values due to their emphasis on the rights of individuals. In contrast, ingroup/loyalty, authority/respect, and purity/sanctity are labeled as binding values, highlighting cohesion within groups through duties and loyalty (Van Leeuwen and Park 2009). The expressions of these moral values play a pivotal role in policy persuasion, as they align closely with the political ideologies that influence individuals' behaviors and attitudes towards policies and governmental issues. This correlation has been explored extensively in political communication studies, revealing that liberals generally prioritize individualizing foundations, while conservatives are more inclined towards binding foundations (Miles 2016).

A similar distinction is also manifested in the discussion over climate change. Much of the political discourse on climate change aligns with individualizing foundations, such as harm and fairness, which may resonate more strongly with political liberals. Democrats typically employ themes of intergenerational equity, sustainability, and moral obligations to nature, leveraging the harm and fairness foundations to advocate for environmental actions (Nisbet 2009; Zhou 2016). Conversely, Republicans are less likely to view pro-environmental actions through a moral lens (Markowitz and Shariff 2012). When they need to emphasize the moral implications of climate change, Republicans may resort to binding foundations like in-group loyalty and deference to authority, reflecting a distinct moral perspective (Markowitz 2012; Markowitz and Shariff 2012).

In addition to the distinction between parties, bipartisan moral values in environmental discussions present a unique case where traditional ideological distinctions may occasionally become blurred. Feinberg and Willer's (2013) study illustrates that when environmental policies are framed in terms of purity/sanctity, the typical ideological differences diminish. This suggests that environmental degradation, viewed as a desecration of the world, can evoke a strong moral response of disgust that transcends political affiliations. Additionally, Wolsko and colleagues (2016) demonstrated that conservatives could be persuaded by pro-environmental appeals in line with their moral values, indicating that the moral divide between political parties may be more flexible than previously believed. Therefore, a thorough examination of the moral values evoked by political elites regarding climate change may provide important insights. Theoretically, this exploration could enhance our understanding of the established relationship between political ideologies and moral values. Practically, it offers guidance on how to strategically articulate different moral concerns to garner broader support. Therefore, we propose the following research question:

RQ1: What moral values do Republican and Democratic members of Congress express in their communications about climate change?

H1: Social media posts (tweets) from Democrats will contain a higher percentage of terms related to individualizing values, whereas tweets from Republicans will feature a greater percentage of terms associated with binding values.

The divergence in moral values between the two parties even extends to their moral language use in discussions of vice (moral violations) versus virtue (morally righteous actions). Studies have identified a pattern where liberals are more inclined to employ terms associated with virtue, whereas conservatives tend to favor terms related to vice. This tendency was highlighted in a comprehensive analysis of U.S. congressional records from 1994 to 2016, revealing that Democrats more frequently used positive (virtue) words, especially in the context of ingroup loyalty, while Republicans were more prone to using negative (vice) words (Takikawa and Sakamoto 2020). Regarding climate change, specifically, climate skeptics were found to prefer vice terms such as "bad" and "false," while still using virtue terms like "truth," "integrity," and "trust" when questioning the validity of scientific research on climate change. In comparison, climate change believers strongly endorsed virtue terms (Diakopoulos et al. 2014). Another study (Fulgoni et al. 2016) found that liberals were more likely to use harm-vice, fairness-virtue and loyalty-vice language when discussing climate change, as compared to conservatives, who used authority-vice more intensively

when denouncing the lack of loyalty and respect for authority. Based on these findings, we propose the following hypothesis for further investigation:

H2: Democrats will use a higher proportion of positive (virtue-related) words in their tweets, whereas Republicans will use a higher proportion of negative (vice-related) words.

These hypotheses aim to explore the nuanced ways in which political parties communicate their moral standings and engage with their target audience through social media, illuminating the strategic use of moral language in political discourse.

1.2 The role of generic frames in congress members' tweets on the climate change

In addition to the moral foundations, generic frames of issues also play a distinct role in shaping public opinion and policy outcomes. Generic frames refer to emphasis frames that make certain aspects of an issue more salient than others, particularly those related to practicality, cost, and self-interest (Slothuus and De Vreese 2010). These frames are instrumental in influencing how political issues are perceived, attributing causes and responsibilities, shaping the perceived importance of issues, and ultimately affecting policy decisions (Cacciatore et al. 2016; McLeod et al. 2022). Their utility lies in their ability to connect complex issues, such as climate change, with everyday concerns and considerations, thereby making them more accessible and relevant to the general public.

Moral foundations of public discourse vary, depending on issues and contexts (e.g., see Graham et al. 2009; Wang and Inbar 2021; Frimer et al. 2017). In the realm of climate change discourse, the ways political parties frame the issue has profound implications for public opinion and policy preferences. Democratic and Republican administrations prioritize different concerns and aspects of the overall climate change issue (Klyza and Sousa 2013). Specifically, Democrats tend to feature testimony from more pro-environment activists and mainstream scientists, whereas Republicans primarily report testimony from speakers in the business and industrial sectors who argue against climate change regulation (Park et al. 2010).

In his 2009 study, Nisbet identified prevalent frameworks used by Democrats and Republicans to discuss climate change, highlighting a strategic division in their narrative approaches. Democratic leaders often employ frames like "*Runaway Science*," which underscore the imminent threat of climate change, and "*Public Health*," spotlighting the negative impacts of climate change on environmental quality and public health. These narratives convey a unified message: the climate crisis affects us all, necessitating collective action against its dire consequences, such as increased infectious diseases, respiratory issues, and extreme weather events.

Conversely, Republican leaders favor frames like "*Scientific Uncertainty*," suggesting climate change is an exaggeration or a manifestation of liberal alarmism, and "*Middle Way*," advocating for bipartisan consensus and balancing economic and environmental interests through clean energy and national security enhancements. Additionally, the "*Economic Consequences*" frame emphasizes that climate policies would result in unfair burden on Americans in international climate agreements if other countries do not take similar actions, and the "*Public Accountability*" frame highlights that the government is prioritizing politics over climate change science. Nisbet's compilation of these framing strategies, drawn from

academic articles, memos, and speeches from the 1990s and 2000s, offers a comprehensive and representative analysis widely cited in subsequent U.S. climate discourse research.

The different framing strategies highlight the divide in partisan interests. This divergence reflects policy inclinations and is exacerbated by political polarization, where frames aligned with political identities resonate more with party constituents. Hence, we ask:

RQ2: How did framing patterns in social media discourse on climate change by Republican and Democratic members evolve from the 111th to the 117th Congresses, reflecting ideological positions?

The two following hypotheses are proposed:

H3: Tweets by Republicans and Democrats will exhibit polarization in their use of generic frames, which are aligned with party interests.

H4: Polarization in climate change framing between tweets from Republicans and Democrats will increase from 111th-114th to 115th −117th Congresses.[2]

Previous research has found that politicians from the minority party tend to use more moral language than those from the majority in the Congress (Wang and Inbar 2021). Specifically, Wang and Inbar (2021) observed that congressional Democrats used more moral language than congressional Republicans across all moral foundations, including binding foundations, between 2016 and 2018. Notably, the differences increased after the election of Donald Trump. Debates exist regarding whether incumbency and party positioning in Congress significantly affect the use of virtue and vice languages. Takikawa and Sakamoto (2020) argue that majority or minority status alone does not result in substantial differences in the use of moral languages. Moreover, other studies have shown that the broader political climate such as elections can amplify the use of moral language. For example, Sterling and Jost (2018) demonstrated that in times of policy shifts, both majority and minority parties might resort to moral language to justify their positions and criticize their opponents.

Environmental issues including climate change were traditionally seen as one of major concerns of the Democratic Party. Democratic congress members' engagement in climate change issues remains stable, while the participation of Republicans tended to fluctuate over time, depending on presidential autonomy (DeGregorio 1998). Fisher and colleagues (2013a) showed that during the 110th Congress when Democrats were in the majority, only 5% Republicans had climate change statements (p. 80). However, this number rose to 24% in the 109th Congress when Republicans controlled both the House and Senate. Meanwhile, between the two congresses, the percentages of climate change statements provided by Democrats changed only slightly.

Although participation of Republican members of Congress in climate change discourse has traditionally lagged, this may be changing. Though they still pursue a different climate agenda than Democrats, they appear more willing to consider climate policies than before.

[2] We treated congress 114th as a "critical juncture" of moral expression because during the the 114th congress, the Republicans took both Houses—with the largest Republican majority since 1931. We also found the moral expression differs significantly before and after the 114th congress. The 114th congress was elected during the final two years of President Obama. From the 115th congress, President Trump and President Biden took office successively. The two time frames analyzed here (111th-114th congress, 115th-117th congress) roughly cover the polarization in Obama and post-Obama eras.

The 110th Congress saw an increasing diversity of actors supporting climate change legislation (Fisher et al. 2013b) and was successful in making efforts to address climate change and energy security through a joint agenda issued by members of both parties (Bang 2010). As a result, a range of hearings on climate change and energy security issues were held and several bills were introduced, many of which attained bipartisan sponsorship (US Senate 2021). In the 116th Congress, Republicans sent out positive signals, as their engagement with the issue continued to evolve.

Finally, by examining moral foundations and generic frames together, we attempt to explore their dynamic interactions. By eliciting moral judgments, moral values can influence individuals' receptiveness to various frames, guiding their attitudes and behaviors towards certain policies (Miles 2016). Therefore, in addition to mapping out patterns of moral language use and generic framing, this study will also examine how Congress members' use moral language to frame climate change, and how moral language is invoked across different generic frames.

RQ3: How does the use of moral language relate to the generic frames advocated by the Congress members, and how does this association vary across parties and evolve over time?

2 Materials and methods

This study examines hypotheses and research questions by analyzing archived tweets from members of the U.S. Congress. Our methodological approach is grounded in Critical Discourse Analysis (CDA), which suggests that the linguistic behavior is shaped by social affiliations and relationships (Fairclough 1995). Within the CDA framework, discourse is conceptualized as a social construct (Wodak and Meyer 2015), and texts are analyzed within their cultural and political contexts (Olausson 2009). This perspective allows linguistic analysis to serve as a powerful tool for examining the ideological processes that influence power dynamics and control mechanisms (Fowler and Kress 1979). In this study, we employed automated computational techniques for linguistic analysis to investigate both thematic (macro-level) and stylistic (micro-level) aspects of discourse (Olausson 2009). To analyze the ideological beliefs of Congress members and their moral foundations expressions and framing strategies in the political persuasion regarding climate change, we focused on their party identification, specifically their affiliation with political parties such as the Democrats or Republicans.

2.1 Data

Tweets were collected using the official Twitter Historical PowerTrack API. The inclusion criteria were as follows: (1) Tweets must contain at least one of five predetermined climate-related keywords (climate, warming, green, energy, pollution); (2) Tweets must have been posted between January 1, 2009, and January 1, 2023, covering the tenure of the 111th to 117th Congresses; (3) Tweets must originate from verified accounts; (4) Only original tweets were considered, excluding replies and retweets; (5) Tweets must be in English. The raw data underwent preprocessing to retain only essential information, such as the creation

time, tweet ID, full text, and user ID. The unit of analysis was individual tweets, and each tweet was timestamped and matched with the corresponding Congress members in office during that period. The final dataset comprises 94,280 tweets, with 60,862 from Democrats and 33,418 from Republicans. A comprehensive overview of the tweets is provided in Table 1.

2.2 Moral foundations assessment

Guided by Moral Foundations Theory (MFT), we applied the extended Moral Foundations Dictionary (eMFD) (Hopp et al. 2021) to assign virtue and vice probability scores to each tweet based on the five moral foundations (i.e., harm/care, fairness/reciprocity, ingroup/loyalty, authority/respect, and purity/sanctity). The eMFD computes the morality values contained in a given text by referencing pre-established dictionaries. The creation of moral foundations dictionaries for textual content analysis has been proposed in previous studies (Graham and Haidt 2012; Frimer et al. 2017). Unlike previous manual selection methods for dictionary construction, the eMFD employs crowd-sourced annotations to ensure the lexicons reflect context-specific human moral judgments. Consequently, the eMFD facilitates the calculation of morality scores for each tweet by comparing tweet content with lexicons in the ten-dimensional moral foundations dictionary.

2.3 Frame analysis

To analyze how Congress members frame climate change on Twitter, we employed a hybrid computer-assisted approach that integrates corpus linguistics techniques with deductive manual classification (Nicholls and Culpepper 2020). This approach began with identifying potential frames and associated framing devices through linguistic analysis (Koteyko et al. 2010). We then proceeded to a deductive stage where elements were coded based on theory-driven frameworks, and subsequently classified the original corpus using frame lexicons (Nicholls and Culpepper 2020). To identify the lexical prominence within the broad corpus, we initially extracted a list of keywords (Sinclair 1991; Baker et al. 2013) by comparing their frequency against the AmE06 reference corpus (Potts and Baker 2012), thereby isolating distinctive lexical elements of Congress members' climate discourse on Twitter.

Using corpus linguistic techniques, we extracted the top 100 keywords, adhering to established norms to offer a representative overview of the corpus (Baker et al. 2013). In the subsequent qualitative categorization, we grouped these top 100 keywords into semantically coherent sets (Van Gorp 2007) and then deductively coded the semantic sets into six established frames of climate change discourse. Next, we conducted collocation analysis,

Table 1 Tweets overview

Congress	Year	Total	Republican	Democrat
111th	2009–2011	3051	1183	1868
112th	2011–2013	7299	4744	2555
113th	2013–2015	9194	4406	4788
114th	2015–2017	9693	4269	5424
115th	2017–2019	13,807	2982	10,825
116th	2019–2021	23,546	5028	18,518
117th	2021–2023	27,690	10,806	16,884
Total	2009–2023	94,280	33,418	60,862

which allowed us to establish terms relevant to the designated frames (Van Gorp 2010). We employed Mutual Information (MI) scores (Church and Hanks 1990) to determine the associative strength of keywords within a five-word proximity, with an MI score threshold of two. This analysis of collocational patterns revealed broader linguistic structures and contributed to the investigation of discourse framing. The resulting dictionaries for each generic frame, consisting of unigram lexicons comprising keywords and respective collocates, served as frame indicators as summarized in Supplementary Information file Table A1.

Lastly, we employed the Linguistic Inquiry and Word Count (LIWC) methodology (Pennebaker et al. 2001) to align tweet content with established frame lexicons. This process involved recording the count of words associated with each frame, calculating the proportional representation of each frame adjusted by tweet length, and determining the tweet's primary frame by identifying the one with the greatest number of matched words. Tweets that did not correspond with any established frame were categorized under "Others." AntConc (Anthony 2022) and Python programming facilitated keyword extraction, collocation analysis, and LIWC calculations, respectively.

Examples of tweets from both parties for each generic frame are provided in Supplementary Information Table A2. These examples were selected based on the highest LIWC scores within each frame, while also accounting for variations among Congress members to ensure a comprehensive overview. It is important to note that the frame analysis captures only the generic frames related to climate change. Within each frame, the partisan discourse can span the spectrum of moral foundations. For instance, the "*Scientific Uncertainty*" frame includes tweets asserting both "climate change is real" and "climate change is a hoax."

3 Results

3.1 Moral foundations expression

We assessed the moral foundations alignment in climate-related tweets by Congress members by employing the eMFD. A higher similarity score indicates a greater degree of moral alignment in the tweets analyzed. For clear demonstration, we calculated the mean similarity score for both individualizing values (care-virtue, care-vice, fairness-virtue, fairness-vice) and binding values (loyalty-virtue, loyalty-vice, authority-virtue, authority-vice, sanctity-virtue, sanctity-vice), as well as separate aggregates for virtue (care-virtue, fairness-virtue, loyalty-virtue, authority-virtue, sanctity-virtue) and vice (care-vice, fairness-vice, loyalty-vice, authority-vice, sanctity-vice) scores. Despite the conceptual debate surrounding the dichotomy of "individualizing versus binding values" (e.g. Graham et al. 2011), these categories are commonly utilized in moral foundations research. We adopted this framework to enhance the interpretability of our findings.

The results challenge H1; the analysis revealed that both Democrats and Republicans predominantly use individualizing values in climate change discourse, with less emphasis on binding values. Contrary to H2, Democratic discourse is more frequently associated with vice-related words, while Republican tweets show a slight preference for virtue-related terms. Figure 1 illustrates these patterns, enhancing our understanding of the moral linguistic strategies in political discourse.

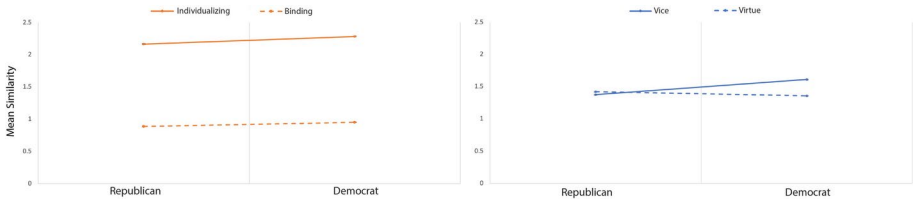

Fig. 1 Comparison of moral foundation expression by republicans and democrats

Table 2 Mixed effect models with random intercepts for congress members and congresses of partisan moral foundation expression

	b	SE	df	t-value	*p*
Individualizing	.15	.01	2592.1	12.1	***
Binding	.10	.01	2431.1	8.84	***
Virtue	-.03	.01	2847.7	−1.97	*
Vice	.22	.02	2960.8	14.73	***

Party affiliation is coded such that 0 = Republican (N = 33,418) and 1 = Democrat (N = 60,862). Degrees of freedom are by Satterthwaite approximation. *p < .05, **p < .01, ***p < .001

We employed mixed-effects models to validate the partisan difference in the expression of moral foundations. To account for nonindependence of tweets authored by the same individual and across different congressional sessions, we included random intercepts for both congress members and congressional sessions in the model. Separate models were built for each of the four types of moral foundation expressions: individualizing, binding, virtue, and vice.

To measure effect sizes, we standardized the moral foundation expression scores, enabling the coefficients to represent the difference in scores between Democrats and Republicans in standard deviations (Lorah 2018). As shown in Table 2, our analyses revealed that Democrats were more likely than Republicans to use language associated with both individualizing values (b = 0.15, t [2592.1] = 12.1, p < 0.001) and binding values (b = 0.10, t [2431.1] = 8.84, p < 0.001). The most pronounced partisan difference emerged in the use of vice-related terms, with Democrats exhibiting a significantly higher frequency (b = 0.22, t [296 0.8] = 14.73, p < 0.001) compared to Republicans. In contrast, Republicans demonstrated a slight yet significant tendency to use virtue-related language (b = −0.03, t [2847.7] = −1.97, p < 0.05). To further substantiate our findings, a detailed analysis of partisan moral foundations, including moral values and moral language use, is provided in Supplementary Information (Table A3-A4).

This pattern indicates that political elites from both parties frequently incorporated individualizing values into their climate change communications, with Democrats showing a stronger inclination to do so. The results confirm that Democrats were more likely than Republicans to utilize both individualizing (b = 0.15) and binding (b = 0.10) values. Furthermore, the difference in effect sizes highlights the practical significance of these rhetorical strategies. Democrats tended to use more negative (vice-related) terminology (b = 0.22), whereas Republicans favored positive (virtue-related) language (b = −0.03). This suggests that Democrats are more likely to employ assertive, negative language to emphasize the urgency of climate action.

3.2 Partisan frame polarization

H3 posits that partisan differences in climate change framing will not only polarize but also align closely with each party's respective political agenda. To test this hypothesis, a series of chi-square tests were performed. To address the nonindependence in the dataset resulting from the clustering of political affiliation, we applied within-cluster resampling using a bootstrapping approach. The use of bootstrapping, which involves sampling with replacement, is well-supported for generating confidence intervals for range statistics when the number of realizations is sufficient to obtain reliable statistics (Davison and Hinkley 1997). Previous studies have validated the use of bootstrap techniques with 1,000 iterations to mitigate the effects of biased variance estimation (Ogunbode et al. 2017). Specifically, we used stratified sampling for the bootstrap method, randomly selecting one observation from each cluster to form a new dataset. This process was repeated 1,000 times to generate multiple new datasets, enabling the estimation of effects. Table 3 details the frequency, count, and significant value of six generic frames utilized by Republicans and Democrats in their climate change Twitter discourse. The analysis reveals that Republicans significantly favored the "*Middle Way*" frame more than Democrats (chi-square (1, N = 20261) = 5166.362, Phi = −0.234, $p < 0.001$), while Democrats showed a stronger preference for the "*Runaway Science*" frame compared to Republicans (chi-square (1, $N = 11889$) = 3885.087, Phi = 0.203, $p < 0.001$). Moreover, the "*Economic Consequence*" frame emerged as the most prevalent overall, accounting for 31.2% of all frames (N = 29,458). Republicans were more likely to use this frame than Democrats, although the effect size was smaller (Phi = −0.116, $p < 0.001$), indicating a less pronounced partisan difference compared to the "*Middle Way*" frame. In summary, a polarized pattern of frame usage emerged: Democrats frequently invoked the "*Runaway Science*" frame (Phi = 0.203) to emphasize the urgency of climate change and advocate for collective action, while Republicans predominantly relied on the "*Middle Way*" frame (Phi = −0.234), focusing on alternative solutions (H3).

H4 states the polarization of partisan climate change framing will increase over the time period we examined. We constructed a log frequency ratio plot for each of six frames, comparing Republican to Democratic usage, with the background color coded in blue and red. The horizontal axis indicates frame preference, with leftward positions denoting Democratic dominance and rightward positions denoting Republican dominance. The size of each frame's circle is proportional to its overall prevalence among the six frames considered. Fig-

Table 3 Chi-square test of partisan frame usage with bootstrap of stratified sampling from congress members and congresses

	Total		Republican		Democrat		Chi-square	Phi	*p*
	%	n	%	n	%	n			
Economic Consequence	31.2	29,458	13.7	12,872	17.6	16,586	1274.6	-.116	***
Middle Way	21.5	20,261	12.2	11,518	9.3	8743	5166.4	-.234	***
Public Health	20.9	19,701	4.9	4602	16	15,099	1590	.13	***
Runaway Science	12.6	11,889	1.2	1175	11.4	10,714	3885.1	.203	***
Scientific Uncertainty	10.5	9867	2	1891	8.5	7976	1276.6	.116	***
Public Accountability	1	902	0.5	435	0.5	467	65.013	-.026	***

*$p < .05$, **$p < .01$, ***$p < .001$

ure 2 illustrates these distributions across two periods: the 111th-114th and the 115th-117th Congresses. Additional chi-square tests with within-cluster resampling were conducted to assess framing shifts across these periods, with results detailed in Supplementary Information, Table A5-A6. The detailed SPSS outputs can be found in the Supplementary Information File 2.

By scrutinizing the frame variation longitudinally, we observe a heightened polarization in the climate change discourse between Democrats and Republicans. During the 115th-117th congresses, both parties frequently employed distinct frames, with Democrats often emphasizing the "*Runaway Science*" frame, while Republicans placed greater emphasis on the "*Middle Way*" frame. In contrast, during the 111th-114th congresses, the two parties adopted more common frames, suggesting a closer alignment in their climate change narratives of both parties during the earlier period. These observations lend empirical support to H4, suggesting a trend toward increased polarization in partisan discourse from the 111th through the 117th congresses, and a growing ideological divide in their respective climate change outlooks.

In our subsequent analysis, we explored the relationship between framing and moral language usage in Congress members' climate change Twitter discourse, focusing on how this association varied across parties and over time. Specifically, we traced the evolution of moral language, categorized as virtue and vice (reflective of positive and negative attitudes) within each climate change frame. Our goal was to quantify changes in the use of moral language over time, considering the relationship with political party affiliation and presidential incumbency. As depicted in Fig. 3, the x-axis represents the corresponding year, while the y-axis denotes the divergence between virtue and vice language—positive values indicating virtue, negative values indicating vice. Overall, we observed a notable variation in the deployment of moral language across different frames, with both Democrats and Republicans tending to use similar levels of virtue and vice languages. Specifically, the

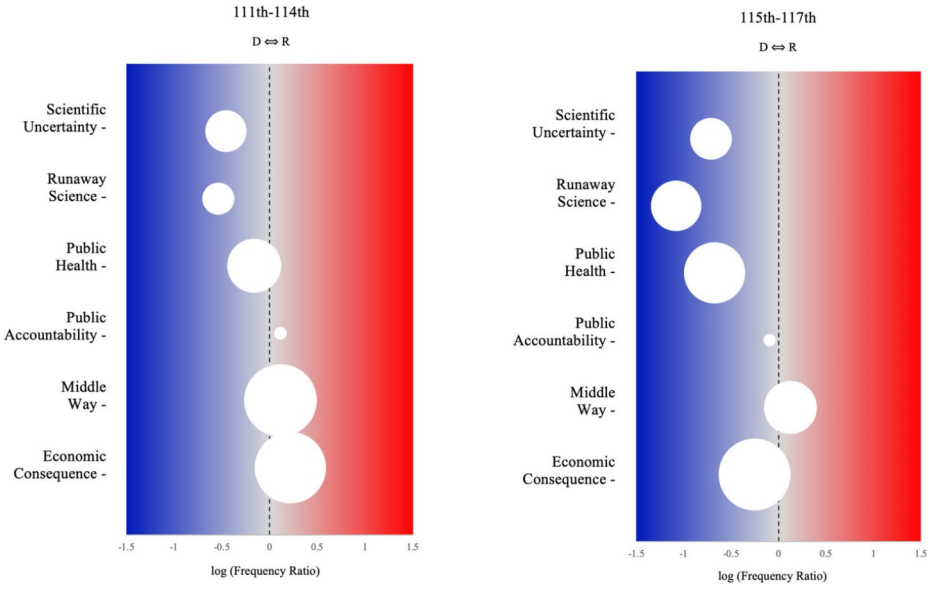

Fig. 2 Comparison of partisan frame usage between 111th-114th and 115th-117th congresses

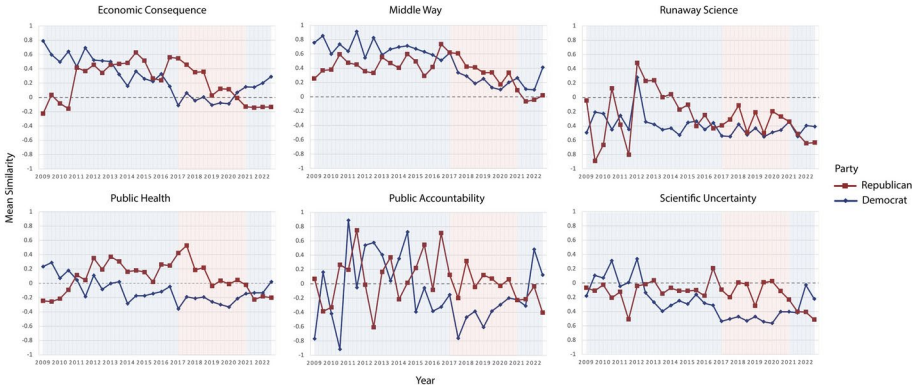

Fig. 3 Partisan virtue and vice moral language usage for each frame over time. *Note.* The x-axis is the corresponding year, the y-axis is the divergence of mean similarity score between virtue and vice. The shaded areas correspond to the political party of the president incumbency, with blue representing the Democratic Party and red representing the Republican Party

application of moral languages varied considerably depending on the frame being considered, regardless of party affiliation. Closer examination revealed that positive attitudes were more prevalent in the "*Middle Way*" frame, while negative attitudes were more pronounced in the "*Runaway Science*" frame for both parties.

To further validate our findings, we conducted hierarchical regression (Supplementary Information Table A7) and double moderation analyses (Supplementary Information Table A8(a) to (d)) to examine the relationships among the six frames, moral values and the the use of moral language in the context of climate change discourse. These analytical methods were designed to evaluate the interplay between framing effects, moral foundations, and political variables in climate change communication. It is important to note that the analysis does not imply causal effects but instead highlights the associations between moral foundations, partisanship, incumbency, and framing.

4 Discussion

This study delved into the 111th to 117th U.S. Congress members' Twitter posts on climate change, uncovering the nuanced relationship between moral expressions, political ideology, and issue framing. Through MFT-based computational analysis, we quantitatively assessed the prevalence of individualizing versus binding values among Congress members and the emphasis on moral language use (i.e. vices vs. virtues) of Democrats and Republicans. Critical discourse analysis was further used to examine the sociolinguistic characteristics of tweets from Congress members, focusing on the partisan polarization of climate change over time, and the relationships between moral foundation profiles, issue frames, and political ideology. This multifaceted approach allows us to uncover the dynamic nature of political communication about climate change, highlighting not just the content but the frequency and context in which these moral expressions were made. Our research investigates the bipartisan preferences for climate change framing and moral language, uncovering the intri-

cate relationship between generic frames and moral foundations. The findings suggest the potential for collective action across the political spectrum.

First, our findings reveal a bipartisan tendency towards individualizing moral values, emphasizing principles of care and fairness. This result challenges the traditional narrative of a stark moral divide between political parties (Haidt and Graham 2007; Graham et al. 2009; Sterling and Jost 2018), suggesting a complex interplay of values. Additionally, Democrats tended to use language associated with vice-related terms, while Republicans preferred virtue-related language. This reflects Democrats' strategic use of moral contagion—where social media can amplify moral outrage messages—to underscore the moral stakes in climate action (Brady and Crockett 2019; Brady et al. 2019). These insights suggest that the rhetorical strategies employed by elites from both parties play a crucial role in framing the climate change, thus having the potential to impact public discourse and to influence policy debates and public opinion on climate change.

Second, both parties appeared to sustain their longstanding positions on climate change, as partisan framing strategies generally appeared to align with the agendas of the respective parties. Democrats continued to advocate for timely climate change action, as they tended to employ the "*Runaway Science*" frame to emphasize the exigency of climate change and the need for immediate action. Meanwhile, Republicans consistently cast doubts on the real impacts of climate change and highlighted the economic costs of pursuing environmental regulations, while opting for the "*Middle Way*" frame to advocate more balanced or reconciliatory solutions.

Third, we identified clear partisan divides in framing climate change through a series of chi-square tests. This polarization has intensified over time, from the 111th to the 117th congresses. Nonetheless, we have also noted areas where common ground might be found. Although Republicans' participation in climate change discourse has traditionally lagged, this may be changing. Specifically, Republicans have also supported pro-environmental development through the "*Middle Way*" approach that advocates clean energy technologies like nuclear, storage, geothermal and carbon capture, as well as legislations for emission mitigation and clean energy.

Fourth, our study shows that the use of moral language (vice vs. virtue) was associated with climate change frames; by contrast, the effects of partisanship and presidential incumbency on this association were relatively minor. For example, a "*Middle Way*" framing may highlight virtue values to encourage participation in collective action, and conversely, vice words may be used in the "*Runaway Science*" frame to deter public engagement (Moser 2007; Wlodarczyk et al. 2017; Markowitz and Shariff 2012). This insight contributes to a deeper understanding of the complexities in the communication strategies employed by Democrats and Republicans, particularly regarding the use of moral language within specific frames.

Fifth, our findings offer a counterpoint to the traditional view that moral values align strictly with partisan ideologies by suggesting that moral considerations in climate discussions are more topic-specific and dynamic than previously thought. As mentioned above, our analysis indicates that specific issue frames are more strongly associated with the use of moral language than party affiliation in the climate change debate, with the influence of party affiliation and presidential incumbency being comparatively minor. This suggests the potential for "ideological symmetry in moral motives," where both parties may frame climate issues in ways that resonate with their moral values—Democrats may frame climate

action as a "sacred" duty, while Republicans might highlight "fairness" in economic development and corporate rights (Frimer et al. 2017, p.33).

Our research suggests that engaging in climate discussions through shared moral values could foster bipartisan dialogue and support for pro-environment policies. The flexibility in moral foundations observed among Democrats and Republicans indicates that moral values, far from being static ideological markers, are dynamically employed by politicians to navigate the complex terrain of political and social identity. This adaptability in moral rhetoric could serve as a potent tool in bridging ideological divides, particularly in the context of the ongoing "culture war." The idea that both parties can frame climate issues in morally resonant ways—whether as a sacred duty to protect the environment or as a matter of fairness in economic development—suggests that moral foundations can offer a pathway to bipartisan consensus and action.

To the extent that Congress members act as opinion leaders and play a key role in correcting misinformation for the public, our findings have important implications for climate change policy and particularly for issue advocacy. Divided elite cues may activate the partisan identity of voters, much to the detriment of issues like climate change where citizens might otherwise find common ground. Morality can function as a strong motivating factor in potentially moving beyond the partisan divide to reach a level of general consensus.

In addition, our findings indicate that when it comes to climate issues, these elites' use of moral language can change over time with the rise and fall in parties' political power. As such, future scholarly endeavors should be wary of the limitations of a static conception of moral claims around climate issues. As Democrats and Republicans may use different moral language in climate discourse in response to different circumstances, a more dynamic view of moral language use will likely be more instructive.

5 Conclusion

This study represents a preliminary effort to apply a combination of computational text analysis and critical discourse analysis to examine moral foundations in climate change discussions. Our findings reveal that, despite the partisan divide surrounding climate change, both Democrats and Republicans converge in moral values and the moral language use related to climate issues. Interestingly, Republicans are moving away from outright denial of climate change towards a more reconcilable "*Middle Way*" to this contentious issue. This shift challenges the prevailing conception that political opinions on climate change in the U.S. are irreconcilably polarized, highlighting several practical approaches for bridging the partisan divide. First, pro-environmental advocates can underscore the new economic opportunities (e.g., in clean energy industries) that climate actions may generate to engage political conservatives. Second, given the shared pattern of moral communication concerning the climate issues, promoting bipartisan dialogue on climate actions is crucial to uniting leaders and citizens across the partisan divide. The findings from this study offer a glimmer of hope for collective action against climate change in the U.S.

It is important to recognize several limitations of the present study. First, our analysis is primarily based on Twitter data due to limitations in data availability. Future studies may incorporate social media data from other platforms for a more comprehensive examination of Congress members' persuasion strategies on climate change. Comparative studies that

examine their online expressions vis-à-vis those of average users will also help map out the broader landscape of climate change discourses as well as the different underlying motivations for content production. Second, our study focuses on the specific issue of climate change. Future research could explore other morally contentious issues or politically divisive topics such as immigration, healthcare, and gun control. Third, there remains room for the further validation of our findings on the interaction between generic frames and moral foundations in other national contexts. We believe that exploring the dynamics underlying moral communication, generic frames, and partisanship for these and other contested issues could provide a more thorough understanding of morally infused persuasion strategies by political elites—and where common ground may be found.

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Data availability The data underlying this article will be shared on reasonable request to the corresponding author.

Declarations

Competing interests The authors report there are no competing interests to declare.

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